"Organizational ambidexterity" and the learning organization: the strategic role of a corporate university

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Abstract

Purpose – Balancing exploration and exploitation is a strategic challenge for technology-based companies striving to successfully implement ambidexterity in rapidly changing markets. This study aims to look at the extent in which corporate universities can be instrumental in the cross-functional deployment of the resources, capabilities and experience needed to achieve organizational ambidexterity.

Design/methodology/approach – This study is the result of a single case study of ZTE University in China. Data from archives, direct observations, and semi-open interviews have been triangulated and analyzed with pattern matching technique.

Findings – This study analyzed the development of capabilities allowing the strategic combinations of exploration and exploitation, and to clearly witness how the corporate university was dynamically linked with those development.

Originality/value – The empirical results offer new insights on the most relevant capabilities for technology-based companies and notably those that are more likely to be exploited through a corporate university.

Keywords Capabilities, Ambidexterity, Innovation, Adaptation, Absorption, Corporate university

Paper type Research paper

Introduction

Understanding the link between learning organizations and innovation gave rise to a rich stream of research (Chiva & Alegre, 2008). Despite the publication of books connecting the phenomena of corporate universities (CU) with the concept of learning organization (Wheeler, 2012; Rademakers, 2014), very few academic publications focused on CU's contribution to organizational learning and innovation within the frame of organizational strategy. Although the CU is often defined as "an educational entity that is a strategic tool designed to assist its parent organization in achieving its mission by conducting activities that cultivate both individual and organizational learning, knowledge and wisdom" (Allen, 2002, p. 3), and despite the fact that research studies have shown how CUs are highly strategic investments for developing learning capabilities (Alagaraja & Li, 2015), this theoretical gap still exists.

This being even more the case when exploration and exploitation strategies are implemented simultaneously, it seems essential to gain a better understanding of the role of

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the CU. For technology-based firms (TBF), strategic simultaneity is common, since they evolve in an extremely dynamic environment that requires continuous exploration and learning new ways to exploit their skills, core competencies and competitive advantages (Floyd & Lane, 2000) and the ability to refine existing products or services (March, 1991; Raisch & Birkinshaw, 2008). They can thus be described as ambidextrous companies which have to simultaneously engage in exploration and exploitation (O'Reilly & Tushman, 2008) which are key capabilities for organizational learning (Asif, 2019). Large and established TBFs typically possess the resources and capabilities necessary to develop and benefit from a strategy of ambidexterity (Voss & Voss, 2013; Wang et al., 2015). Indeed, TBFs are in some ways associated with the development of ambidexterity because they have both the need and ability to simultaneously exploit and explore and balance these capabilities (Chandrasekaran, Linderman, & Schroeder, 2012). Considering the heavy investment required to create CUs in 4000 TBFs worldwide (Kolo, Strack, Cavat, Torres, & Bhalla, 2013), the dearth of studies on their role in exploration and exploitation is surprising (Lissillour & Rodríguez-Escobar, 2021).

Thus, if highly developed CUs do support strategic decisions leading to effective exploration and exploitation, then this support should be observable in ambidextrous organizations. However, this hypothesis is not backed by any previous studies. Therefore, the present research objective is to provide evidence of the role of CUs role within an ambidextrous company and to analyze its main actions. Thus, this paper focusses on the following research question: how does CU contribute to the development of ambidexterity? Specifically, we draw on a case study of the company ZTE, the strategic behavior of which displays the conceptual characteristics of an ambidextrous organization and which has been developing a CU for over two decades. The following sections of this research focus on conceptually explaining, in terms of capability, the scope a CU can have in the deployment of an ambidextrous strategy. Subsequently, drawing on the case study, we determine patterns coinciding with the factors that constitute such capabilities, and we finally conclude by setting out theoretical contributions and practical implications.

Exploration and exploitation of capabilities

Conceptual development of ambidexterity

Several decades have passed because Chakravarthy (1982) assured that organizations had to adapt to changes in the environment, and that within this survival effort, a process of adjustment of the business structure and strategy was estimated. While the validity of these arguments is not contested, the strategic dilemma remains in terms of the orientation towards the exploration or exploitation of business units and to the complexity of balancing these two approaches and combining them alternatively. In the configuration of their objectives, companies may be inclined towards the search, discovery and experimentation of new opportunities (exploration strategies) or to implement and enhance the efficiency of their current resources (exploitation strategies) (March, 1991). Following Lavie, Stettner, & Tushman (2010) advice, this study conceptually relates exploration and exploitation to March's (1991) definitions. Exploitation is thus defined in terms of "refinement, choice, production, efficiency, selection, implementation and execution," while exploitation involves "search, variation, risk-taking, experimentation, play, flexibility, discovery and innovation" (March, 1991, p. 71).

The differences between each approach are significant and have a strong but conditioned imprint on organizational dynamics such as the corporate learning. Sitkin, Sutcliffe, & Schroeder (1994) simplified this influence arguing that organizations that privilege exploitation develop learning activities involving the use of resources that the company already has, while those that privilege exploration develop learning activities that lead to the addition of new resources. However, "effective selection among forms, routines or practices is essential to survival, but so also is the generation of new alternative practices, particularly in a changing environment" (March, 1991, p. 72). This argument is the one that companies use to support their decision to bet on both approaches at the same time and thereby configure, achieve, assign, adapt and allocate the resources necessary for this ambidexterity, which would also imply the ability to simultaneously work on the alignment and adaptability across an entire business unit (Gibson & Birkinshaw, 2004).

Ambidexterity implies an effective communication of the vision and values throughout the organization and that the organization successfully develops the capabilities and technologies needed to compete in new markets and thus survive in a changing environment (O'Reilly & Tushman, 2008). This requires the collective development of discipline and efforts toward organizational trust and a more cooperative environment (Günsel, Altındağ, Keçeli, Kitapçı, & Hızıroğlu, 2018). Ambidexterity develops distinct logics nested at multiple levels in the organization (Birkinshaw & Gupta, 2013). Consequently, the logics embodied in exploration and exploitation include contradictions which often create tensions within the organization (He & Wong, 2004).

However, Ebben & Johnson (2005) have found that companies with strategies that focus simultaneously on efficiency and flexibility underperformed compared to those with a single strategic focus. Likewise, ambidexterity is not necessarily positively linked with the achievement of objectives, notably in case of limited access to resources (Atuahene-Gima, 2005; Lin, Yang, & Demirkan, 2007) and low technological industry (Uotila, Maula, Keil, & Zahra, 2009). Ambidexterity may undermine organizational performance because it traditionally translates into establishing a division between exploring and exploiting units, instead of ensuring that units efficiently alternate between these two approaches (Lavie et al., 2010). Thus, companies that choose to combine exploration and exploitation in particular organizational domains and achieve a balance between these domains, may perform better on the market. This perspective stresses the importance of the internal and external adjustments necessary to combine exploration and exploitation, which imply the creation, development and strengthening of different capabilities.

Adaptation, absorption and innovation

We will define capabilities "as a set of actions (or routines) taken by senior management that permit the enterprise to identify opportunities and threats and reconfigure assets (people, organizational architectures and resources) to adapt to these" (O'Reilly & Tushman, 2008).

Thus, if capabilities exert their catalytic role and if they are correctly combined; they can mediate ambidexterity and its outcomes (Kristal, Huang, & Roth, 2010). Moreover, these dynamic capabilities can help organizations identify opportunities and threats and reconfigure resources to adapt to change (O'Reilly & Tushman, 2008). Indeed, key capabilities are configured to exploit existing assets and positions and explore new technologies and markets to adapt the organization to a changing competitive environment (Teece & Pisano, 1994; Teece, Pisano, & Shuen, 1997). In other words, companies must develop sufficient capabilities to be able to strategically combine exploitation and exploration and thus successfully reconfiguring resources (Wang & Ahmed, 2007). Therefore, for the development of ambidexterity, there must be an adjustment of strategic capabilities by making a continuous review of its objectives (Chakravarthy, 1982), its incentive policies (Boyd & Salamin, 2001) and the improvement of internal technological, organizational and managerial processes (Teece et al., 1997).

Three main components of the capabilities are involved in the exploitation and exploration strategy, namely, adaptation, absorption and innovation (Wang & Ahmed, 2007).

The adaptation focuses on identifying and capitalizing on emerging market opportunities including the organizational capability and material capability (Chakravarthy, 1982). The first is defined as the information processing capability of the company. The second is the capability to gather resources closely linked with the strategy, while implementing mechanisms for the regulation of these resources. In this research, we limit material capability to the ability to obtain, produce and retain resources, as we do not investigate the quantity possessed or required for specific activities which would lead us to other interpretations that come out of the context of this paper. Because adaptability has an intimate relationship with the strategy, it involves a broad spectrum of possible dimensions such as both individual and collective adaptation a leadership that can alternatively be distributed (Fu, Liu, & Liao, 2018) and visionary (Rao-Nicholson, Khan, Akhtar, & Merchant, 2016). New management methods boost adaptation potential, namely, explicit feedback loops (Voss & Voss, 2013) and continuous monitoring (Kaur, Gupta, Singh, & Perano, 2019). Further elements contribute to efficiently deploy adaptation such as continuous learning experience (Parent, Roy, & St-Jacques, 2007) and critical thinking capability (Khan & Mir, 2019).

Prior research described absorption as an external learning capability (Camisón & Forés, 2010; Jansen, Van den Bosch, & Volberda, 2005a) that allows organizations "to recognize, assimilate and apply externally held knowledge" (Ferreras-Méndez, Newell, Fernández-Mesa, & Alegre, 2015, p. 1). Knowledge acquired externally is used to create value in four steps, namely acquisition, assimilation, transformation, exploitation (Zahra & George, 2002). Exploitation perfects and extends current knowledge, seeking greater efficiency and improvements, while exploration implies the development of new knowledge to experiment and promote variation and novelty (Atuahene-Gima, 2005). This capability is usually present reliable, flexible and administratively structure organizations that evolve in knowledge-based changing environments (Parent et al., 2007). This is typically the case of TBF (Wang, Chang, & Shen, 2015).

Organizational factors and analysis grid

Innovation is perhaps one of the most studied because it focuses on understanding the variables that lead companies to successfully develop new products and/or markets (Saunila & Ukko, 2012). For TBF s that typically operate in high-tech markets, the ability to innovate is more than critical and those that achieve greater deployment of these capabilities can be expected to generate higher profits than those that are not innovative (Balkin, Gideon, & Gomez-Mejia, 2000). Previous research studies have established listing of organizational capabilities that enable innovation (Iddris, 2019; Lau & Lo, 2019). In this paper, rather than seeking to re-reviewing the capabilities found in the literature, we have sought to create 16 capabilities that allow for a synthetic understanding of the role of CU in the deployment of incremental and radical innovation in a company. Table 1 (in supplementary materials) summarizes the capabilities identified in the literature that can contribute to the proper deployment of ambidexterity as adaptation (code AD), absorption (code AB) and innovation (code IN).

Inspired by the morphological analysis framework of dynamic capabilities (Sunder, Vijaya, & Marathe, 2019), we suggest a logical grouping of the capabilities according to their salient attributes into seven factors. *Organizational antecedents* affect the level of organizational ambidexterity such as the extent of decentralization and social density

Factor	Code Type	N°	Dimension	Reference
Organizational antecedents	AD	1	Organizational capacity	Chakravarthy (1982)
	IN	3	Management capability	Zawislak et al. (2012)
	IN	5	Strategic planning capability	Lau & Lo (2019)
	IN	6	Organizational capability	Yam et al. (2011)
Organizational resources	AD	2	Material capability	Chakravarthy (1982)
	AD	7	Explicit feedback loops	Voss & Voss (2013)
	AB	2	Acquisition capability	Yli-Renko et al. (2001)
	AD	3	Experience	Parent et al. (2007)
Leadership	AD	4	Visionary leadership	Rao-Nicholson et al. (2016)
	IN	10	Cultural and leadership context	O'Reilly & Tushman (2008)
	AD	5	Critical thinking capability	Khan & Mir (2019)
	AD	6	distributed leadership	Fu et al. (2018)
Organizational change	AB	4	Exploitation capability	Yli-Renko et al. (2001)
	IN	2	Innovation decision capability	Wang et al. (2015)
Technological change	IN	4	R&D capability	Yam et al. (2011)
	IN	7	Technological capability	Figueiredo & Brito (2012)
	AB	3	Transformation capability	Zahra & George (2002)
Exogenous	AD	8	Continuous monitoring	Kaur et al. (2019)
	IN	8	Collaboration with external parties	Börjesson & Elmquist (2011)
	IN	9	Communication of project ideas	Börjesson & Elmquist (2011)
Organizational learning	IN	1	Learning capability	Yam et al. (2011)
	AB	1	Assimilation capacity	Mardi et al. (2018)
	IN	12	Experimentation with new methods	Börjesson & Elmquist (2011)
	IN	13	Training	Whittaker et al. (2016)
	IN	11	Know-how development	Saunila & Ukko (2012)
	IN	14	Creativity and idea management	Lawson & Samson (2001)
	IN	15	Intra-organisational knowledge sharing	Calantone et al. (2002)
	IN	16	Transaction capability	Zawislak et al. (2012)

Notes: AD = Adaptation capability; AB = Absorption capability; IN = Innovation capability

Table 1. Organizational capacity

(Jansen et al., 2005). Organizational capability (Chakravarthy, 1982) has implication in terms of exploitation, notably strategic planning capability (Lau & Lo, 2019) and communication of project ideas (Börjesson & Elmquist, 2011).

Ambidexterity implies appropriately dealing with *internal and external organizational resources* which requires the capability to acquire them (Yli-Renko, Autio, & Sapienza, 2001) before adapting, integrating and reconfiguring them. Organizational resources include various capabilities such as material capability (Chakravarthy, 1982), RandD capability (Yam et al., 2011) and Technological capability (Figueiredo & Brito, 2012). Prior research has shown that *leadership* is a factor that includes behaviors that rely on specific organizational mechanisms to effectively promote ambidexterity (Baškarada, Watson, & Cromarty, 2016). Leadership style such as visionary leadership (Rao-Nicholson et al., 2016) and distributed leadership (Fu et al., 2018) have direct implications in terms of adaptation capability.

Organizational change is an important factor because it the context in which organizational ambidexterity develops, and it develops even so better that companies design and implement a robust change management program (Mitra, Gaur, & Giacosa, 2019).

Exogenous factors include capabilities like collaboration with external parties such as suppliers and customers (Börjesson & Elmquist, 2011) but also includes capabilities to adapt to regulatory and legal factors (Sunder et al., 2019). Different types of *organizational learning* play a role in facilitating organizational ambidexterity throughout the innovation process (Seidle, 2018) and have implications in terms of assimilation capability (Mardi, Arief, Furinto, & Kumaradjaja, 2018) and transaction capability (Zawislak, Cherubini Alves, Tello-Gamarra, Barbieux, & Reichert, 2012). Organizational learning constitutes a capability that plays a moderating role on employee satisfaction, entrepreneurial orientation and firm performance (Chiva & Alegre, 2008; Chiva & Alegre, 2009; Alegre & Chiva, 2013; Fernández-Mesa & Alegre, 2015). Learning involves additional capabilities such as intra-organizational knowledge sharing (Calantone, Cavusgil, & Zhao, 2002), training (Whittaker, Fath, and Fiedler, 2016), creativity and idea management (Lawson & Samson, 2001). Technological factors include the role of information technology to push innovations in business processes (Jansen et al., 2005), but it also includes the extent in which RandD capability (Yam et al., 2011) are systemically pursued and converted into actionable deliverable thanks to the firm's transformation capability (Zahra & George, 2002). These seven factors suggest inter-group relationships between these capabilities. Indeed, a complex orchestration of skills and capabilities create valuable, rare, inimitable and non-substitutable features based on the competitive advantage of each unit and subsequently of the entire company (Jansen, Simsek, & Cao, 2012).

Corporate university

Organizational context

Prior studies characterized CU according to three main functions (Fresina, 1997). The first is typically to ensure the delivery of individual learning programs with a focus on processing and transferring knowledge. This first function allows a gradual monopoly on training delivery that contributes to its institutionalization into the company. The second function is to manage change and focuses on preparing employees and connecting them to the change processes. Finally, the third function is to engage in strategy making and shape the "strategic renewal" of the organization. Provision of internal training is then defined as a facilitator to solve specific problems and improve the performance of an entire organization.

Another way to understand the scope of a CU is through its development phases. Prior research studies propose three main phases (Jansink, Kwakman, & Streumer, 2005). The first is the operational phase, during which the organization's training offer is organized and centralized. The second is the tactical phase, during which focuses on enhancing knowledge dissemination and exchange. The third phase is the strategic phase, during which efforts are made to research and build strategically relevant knowledge. Moreover, CUs are described as a strategic investment aimed at improving the corporate culture, allowing both individual and organizational learning and enriching the corporate knowledge (Allen, 2002). Given that the conception, orientation and use of CUs may ultimately differ from organization to organization, its performance may vary ostensibly (Parshakov & Shakina, 2018). The complex coupling between its functions, objectives and structures with the specific features of the company (productive and commercial activity, cultural and organizational dynamics and interactions with the environment) positions the CU as a more necessary and competent unit than traditional universities (Blass, 2005; Nixon & Helms, 2002). Large TBF's CUs gave rise to advanced systems of knowledge management which are instrumental for their organizational growth, because training programs *per se* do not ensure the effective transfer and application of knowledge (Rahman, Ng, Sambasivan, & Wong, 2013) and drive knowledge innovation (Rademakers, 2005). Indeed, CU that evolve in very dynamic markets have to do more than just update employees via training programs (Wang, 2014) to provide value to their organization.

For many companies, the CU is a consolidated strategic asset in terms of the organization and human resources management (Patrucco, Pellizzoni, & Buganza, 2017) and should thus be understood as an investment aimed at disseminating the corporate culture and enabling both individual and organizational learning, while also enriching corporate knowledge (Allen, 2002). Thus, world-class universities are characterized by a comprehensive learning infrastructure which formalizes the corporate culture and are instrumental in transforming companies into real learning organizations (Eccles, 2004). However, this ideal is often not reached, and the scope of some CUs is limited to the operational delivery of training to employees (Jansink et al., 2005). This limited development of CUs is often likely because of a lack of interest and long-term planning on the part of their organization, as leaders are sometimes hostile to investing in intangible units which indirectly maximize future performance (Parshakov and Shakina, 2018).

Capability approach

Previous research has shown that a key success factor for CU is to be strategically aligned with the corporate strategy and corporate structure to boost organizational performance (Lissillour, Rodríguez-Escobar, & Wang, 2020). Thus, the alignment effort should supposedly be higher in the case of an ambidextrous organization because performance arguably depends on the congruence of the strategic combinations with the market (Voss & Voss, 2013) and the organization. CU operate across or within functional domains as they support functional departments with learning services and thus contribute to facilitating organizational change (Wang, Li, Qiao, & Sun, 2010). But exploration and exploitation require distinct learning models (O'Reilly & Tushman, 2008). Consequently, a CU could take on the challenge of supporting a strategic balance and of overcoming the substantial divide between routines and learning, which are likely to hinder the combination of exploration and exploitation (Gupta, Smith, & Shalley, 2006).

In practice, many companies opt for this managerial balancing act between exploration and exploitation, but for Technology-Based Firms (TBF), the balance between these two approaches is almost more an obligation than an option. Indeed, "the more uncertain the situation, the more an organization will need flexibility to complement planning" (Kouropalatis, Hughes, and Morgan, 2012). Thus, innovative organizations naturally tend to incorporate strategic units that produce knowledge (Cui, Ye, Teo, and Li, 2015). Indeed, knowledge management requires the establishment of a strong infrastructure, which encompasses fully articulated people, technology and procedures (Cepeda and Vera, 2007). However, while there is a certain consensus on the need for balance between resource optimization on the one hand and learning and innovation on the other, there is a lack of information regarding how to reach this equilibrium (Gupta et al., 2006). One possible response could be to generate dynamic capabilities because they "affect stages of the adoptive management innovation process; from initiation through to implementation." (Lin, Su, and Higgins, 2016, p. 1). Indeed, these capabilities are manifested in functional activities such as RandD, marketing and distribution, technology and operations applied to the business activity (Helfat, 2002). Thus, TBFs which have a CU should strive to create specific dynamic capabilities. To fulfill their role as an organizational learning facilitator, CUs must consider the development and maintenance of central processes, such as knowledge systems and processes, networks and associations, collective and individual learning processes (Prince and Stewart, 2002).

Methodology

This research used a qualitative and exploratory approach to observe a contemporary complex phenomenon embedded in a real-life context (Eisenhardt, 1989; Yin, 2003). The choice of a single case study methodology is justified because the case has not been accessible to researchers before and could be observed longitudinally (Yin, 2003). Moreover, this study involved practitioners, namely the dean, prior dean and several managers from ZTE and claims to generate practical contributions, therefore the case study is a method well-suited to the present research purpose (Amabile et al., 2001).

Case selection and data collection

The selection of our case was led by our research objective, namely, to analyze the role of CU in TBF in the development of ambidexterity. To perform a longitudinal study, we looked for a TBF with an already consolidated CU. The first contact with ZTE Corporation was in 2015, via a program manager who gave us access, which resulted in an initial analysis of the strategic management of CUs. It should be noted that it is a benchmark company in the telecommunications and information technology sector worldwide, managing to position itself as one of the most important companies in China in this industry and the fifth in the world in the past ten years. In addition, the company has created its CU nearly twenty years ago and is today well-defined and perfectly integrated into the corporate culture, physical structure and strategy.

The data collection and focused on three types of data (see Table 2 in supplementary materials), namely archives of ten years of news concerning the CU, direct observations and five semi-open interviews based on an interview guideline constructed according to the theoretical framework that have been recorded and transcribed. The design of the interview guideline followed the main dimensions identified in the theoretical framework. The semi-open interviews were carried out with the current and former deans of ZTE CU, a program director and a marketing manager. Relying on case study research principles rather than statistical sampling, the choice of respondents followed a theoretical sampling principle (Glaser and Strauss, 1967) to match theoretical patterns rather than replicating previous cases. We selected interviewees for their ability to provide useful information on concepts relevant to the emerging theory or which provided additional insight into relationships among concepts (Eisenhardt and Graebner, 2007). A case database was built with the private documentation that was accessed, information published on the corporate website, specialized and commercial press, field notes, as well as individual interviews.

Sources	Quantity	Description
Interviews	5	Semi-structured interviews conducted between 2017 and 2018 based on an interview guide, where in some cases questions were refocused and new additional questions were asked for more clarity
Dean 2014	1	
Marketing manager	1	
Program director	2	
Dean 2018	1	
Various documents	40	Business documents, (brochures and manuals issued by the CU), management documents (activity manuals, organization principles, strategic plan and process manual)
Files	27	Directors' speeches and events archives
Media publication	35	Institutional website (product information, service and current news) Other websites (general information on the company and its competitors)

Table 2.
Data collection

Data analysis
This diversified set of multiple data sources allowed for triangulation and thus enhanced the reliability of the results (Stavros and Westberg, 2009). Likewise, given the complexity of the object under study and the difficulty of gathering quantitative data, the qualitative method turns out to be the most appropriate (Woodside, 2010), notably when capabilities are discussed and analyzed (Mohamud and Sarpong, 2016).

The data analysis occurred in four steps. The first step focused on organizing chronologically the case database from the past ten years so that we could describe the development of the CU. Then the database was coded according to the dimensions identified in the theoretical framework. The qualitative case study thus focuses on interpreting and crossing data from diverse sources to enable triangulation to gain a better understanding of the capabilities that ZTE developed along with the development of the company's CU.

We used the pattern matching technique, which consisted of identifying and comparing patterns contrasted by the literature or hypothetically predicted patterns related to the types of capabilities that facilitate the implementation of exploration and exploitation strategies, thus comparing them with the data and evidence collected through interviews and secondary sources of information, to understand why and how decisions are made and literally fits with "exploratory and inductive research [which] focuses not on outcomes or results of human behavior, but directly on the causal mechanisms that underlie and produce social phenomena" (Reiter, 2017).

Case analysis: ZTE University and the development of capabilities
Ambidexterity is itself a dynamic capability that generate new capabilities that maintain a balance between exploration and exploitation strategies as a basis for their adaptation in a volatile environment (O'Reilly & Tushman, 2008). Thus, complex connections develop between capabilities, such as decentralization, differentiation, cultural consolidation, leadership and commitment, internal and external growth. The accumulation of strategic routines sustains the capacity of the company to cope with rapidly and continuously changing markets (Teece and Pisano, 1994). The CU is thus a strategic asset of the company:

> The corporate university of ZTE has always been a strategic institution that is highly valued by our top leader, who devoted a lot of time and effort in engaging in relevant work. He is just like the CEO of GE – Jack Welch – who also attached great importance to GE Crotonville (Dean, 2014, p. 4).

The events identified in this study helps us explain chronologically how they directly generated capabilities that indirectly influenced the development of other capabilities.

In the first period described (The origin of the ZTE university), the first measures were to adapt quickly to current strategic demands which were mainly determined by the consolidation and business growth of the company. At that time, ZTE recognized that it is necessary to improve the service portfolio to match the level of increasingly innovative products. The company has a clear absorption capacity (Cohen and Levinthal, 1990), recognizes the value of information that comes from the market and is able to assimilate and seek solutions to be applied for commercial purposes. Thus, through the CU, ZTE intends to strengthen all the sections of its product value chain, which in turn meant improving the quality of services, training of the technical and administrative operation. Consequently, the company brought additional resources to strengthen the structure of the CU, thus enhancing its infrastructure and the qualification of employees. The incorporation and development of new resources required organizational and management capabilities, such as visionary thinking, leadership and assimilation and transformation of resources.

In the first period, a gradual transition occurred that shifted the focus of trainings from external clients to internal clients. This transition is fundamentally because of the fact that the company was not sufficiently prepared to supply and adapt quickly to the new demands. Consequently, it was essential for the CU to improve the training resources for internal clients to continue innovating and providing top services. To respond to external changes, the company must be oriented in the adaptation, renewal, update of resources and reconfiguration of central capabilities (Wang and Ahmed, 2007), which is precisely what ZTE did using the CU as the engine for change. Thus, the CU created new departments (such as the PMO), incorporated new technical and human resources and developed new organizational and learning capabilities to continue growing and improving its contribution to the company. ZTE University gradually:

> emerged as an independent institution. As the company became bigger and more complicated, the CU was allowed more flexibility and opportunities to make its own decisions. From this time on, the university was able to operate its budget independently and to generate profit (Dean, 2014, p. 4).

Among the new functions assigned to the CU, the generation of technical documentation particularly challenging because it involved technical precision and a communication language close and accessible to different types of clients. To fulfill this new function, the CU worked to integrate the capabilities and knowledge of the RandD and commercial departments into its own structure. This absorption capacity dominates the organizational adjustment (Senivongse, Bennet, and Mariano, 2019), but there is also a continuous capacity for innovation and adaptation that allows the dynamic adjustment of resources and existing capabilities to new market demands. Because then, the documentation service has proven to be a very strategic asset for the company, as it contributes to improve both the operational efficiency and customer satisfaction. This new capacity enhanced the status and influence of the CU within ZTE:

> Our documentation includes product documentation and documents for engineering and learning purposes, etc., which functions as knowledge management (Dean, 2014, p. 3).

As ZTE engaged in large-scale internationalization in 2006 (Second period), talent development, knowledge retention and dissemination processes became crucial. When internationalizing, according to Lavie et al. (2010) "organizations benefit from access to remote and diverse resources, yet face challenges because of increasing cultural, institutional, economic, and geographical distance". Therefore, when the company ventures into new countries and establishes different collaboration and commercial agreements, maintaining the service portfolio is challenging because of logistic and linguistic issues. This case confirms that human and organizational capabilities, as well as innovation and commercial capacities are key for internationalization (Camisón and Villar, 2009). Consequently, the CU created 12 training centers abroad, which had the capacity to offer services similarly as at headquarters, but also developed programs to be delivered online. Moreover, new services such as language training for commercial and administrative staffs required adapting the infrastructure and organizational structure. The shortcomings that were detected at the beginning were gradually eliminated, and the CU could thus reduce uncertainty and improve the efficiency and scope of the company's internationalization process. It also allowed the deployment of the capabilities to absorb knowledge from the external environment and convert them into internal knowledge (Cohen and Levinthal, 1990), thus raising new capabilities instrumental for adaptation and innovation. The CU became a tool with which to disseminate the values of ZTE and a channel for its corporate social responsibility:

> [. . .] we can leverage the platform provided by the university to fulfil our corporate social responsibilities, which include providing certain social services [to the local population]. [. . .] We inform local universities, clients and social institutions to let them know that we offer free technical training for ordinary people. Communication technologies are still lagging behind in many countries. Even many local university teachers do not have enough knowledge on communication technologies, but we can help them by providing training. For instance, some countries have no idea about 5G, so we have arranged some courses on that topic (Dean, 2014, p. 6).

The CU became a place wherein new management methods could be assimilated by the company:

> In the past, when we talked about project management, we traditionally referred to sales, R and D, engineering departments, etc. But among our functional departments, it was the university that first integrated this management mechanism into its system. [. . .] The second leader of the CU was appointed as the Executive Vice President of ZTE due to his contribution in the creation of the Project Management Office (Dean, 2014, p. 2).

In 2007, the beginning of the third period, different measures were taken to connect training programs with strategic processes, that is, to put training at the service of the strategy, not only in activities to articulate its implementation, but also to facilitate the design, communication, acceptance and cultural integration of each organizational change. The CU, therefore, had to face different challenges such as creating training programs that are useful, targeted and sufficiently flexible to adapt to changes or strategic context:

> [. . .] the university itself assumes an independent role [. . .]: we can understand it as a consulting agency (Dean, 2014, p. 8).

Consequently, the CU adjusted content and sequence of activities and directly involved high-level managers into the leadership programs, to ensure constructive exchanges articulated around different strategic propositions.

Initially, the Leadership Development Program targeted top managers and delivered general information about the results of the company and imparted on current trends in leadership and management. However, it become a permanent forum that serves as a space for discussion about strategic planning. Throughout its history, the CU has remained closely linked to the needs and expectations of the commercial departments, but this last period led to a change in its strategic integration which enable contributions to the elaboration of the global strategy.

ZTE University has been able to adapt and improve its structure and reconfigure its capabilities depending on the corporate strategies. The CU developed the capacity for strategic alignment (Lissillour et al., 2020) and simultaneous adaptability, very close to the concept of contextual ambidexterity (Gibson and Birkinshaw, 2004).

Table A1 summarizes key events identified in the study that have motivated the deployment of specific capabilities that directly and indirectly enabled the development of the company's strategic ambidexterity. Each of the elements collected is chronologically decisive for the construction and strategic development of the company and integration of the CU in this process. Moreover, Table A1 also illustrates the dynamism of the cluster of events" programmed by the company management and articulated by the CU. These "events" occur in conditions of high systemic entropy, characterized by significant fluctuations in terms of motivation, creativity and efforts towards continuous improvement. This is indeed an example of how dynamic capacities are created in practice, which is not as straightforward as the theory might predict.

Discussion and conclusion

This study analyzed the operationalization of strategic ambidexterity through the deployment and operation of a CU from the perspective of dynamic capabilities. The case study method with multiple data sources adds to the robustness to the results. Thus, there were two methodological challenges. The first was related to determining which companies could be considered strategically ambidextrous and the second involved finding an organization that had a well-developed CU. Hence, a TBF such as ZTE proved to be the perfect case study for measuring the degree of penetration and contribution of its CU in the strategic deployment of simultaneous exploration and exploitation.

Theoretical contribution

The first theoretical contribution is the development of an analysis grid composed of capabilities classified in combinable and combined factors. These factors do not necessarily manifest themselves operationally in term of adaptation, absorption and innovation as configured in the literature. Indeed, depending on the strategic development, companies switch from one factor (organizational antecedents, resources, change, learning, technology and leadership) to another. While doing so, companies develop learning processes that detect combined capabilities and reconfigure tangible and intangible assets to maintain their competitive advantage (Helfat, 2002; Qiao, 2009). Thus, the perspective of ambidexterity is consistent with the literature on knowledge acquisition, although it does not focus on the practical means to develop ambidexterity. Consequently, exploring CU as an instrument to develop ambidexterity is useful and even novel because there are no previous studies that examined this relationship.

The present findings show that the intervention of a CU in strategic development can take various forms, mainly when the CU acts as a business unit which can positively contribute to strategy implementation (Rhéaume and Gardoni, 2015). Such intervention, which is enacted through the consolidation and deployment of dynamic capabilities of different orders, can in turn generate value through creativity and the renewal of other capabilities (Kogut and Zander, 1992) including socialization (Moh'd Al-adaileh, Dahou, and Hacini, 2012). More especially for a company that has an ambidextrous strategic trajectory, which is usual for TBFs, the support of the CU is instrumental, because simultaneous exploration and exploitation represent efforts to develop adaptive, absorptive and innovative capabilities. The results obtained also reveal patterns of how these capabilities are deployed through key events, such as transformation, use of knowledge, training, knowledge exchange, learning, development of know-how, cultural development and leadership. These have been present in many events, acting as integrators of exploration and strategic exploitation. Nevertheless, the results show that technology and RandD capability have had a lower deployment in the activities carried out by the CU. This can be explained, on the one hand, because the TBFs have specialized departments in charge of these capabilities and, contrary to what one might think at first, the CUs are not so necessary in this area.

Similarly, we found that the CU served from the beginning as a catalyst which convert strategy into action and flexibly implement learning mechanisms. Maintaining a balance between strategic and operational objectives, the CU facilitated the creation of a corporate culture that supports the achievement of the strategic objectives of the company. This feature can be especially useful to facilitate change in culturally plural organizations (Lissillour and Wang, 2021).

Practical implications

This research studies have practical implications for change managers from large organizations that must respond immediately to permanent changes in the environment. Creating and implementing a CU can facilitate the configuration of capabilities that create a sustainable competitive advantage, as long as there is the strategic alignment with the organization (Anderson, 2009; Lissillour et al., 2020). Likewise, companies can simultaneously dedicate themselves to exploiting basic capabilities for the benefit of the organization and exploring new capabilities in more creative ways inside and outside the organization. However, they must be aware that they need commitment and collective responsibility from all stakeholders, a well-defined structure, cultural consensus, combined with an active and participatory leadership. In short, boosting adaptation and absorption capacities to seek and connect with external learning sources (Love, Roper, and Vahter, 2014) ultimately facilitates the creation and consolidation of innovation capacities for the entire organization. Hence, raising the CU's level of service is likely to make these innovation capacities more relevant and consistent with the corporate strategy, because CU services effectively facilitate the operationalization of the company's strategic intention (Smith, Jayaram, Ponsignon, and Wolter, 2019).

Limitations and avenues for future research

This research has methodological limitations, because only one case study was conducted, which may limit empirical generalizability. Although this study uses multiple research instruments such as direct observation, interviews and documents, to understand the development of the CU and its contribution to strategic orientation, there was no access to specific data on operational or financial results that could establish a capacity-building link with the organization's performance. Although the public information demonstrates the financial health and growth of the company, future studies may look deeper into this link.

Finally, we suggest that future research consider other characteristics of CU that may no longer indicate a role of mediation, but rather of moderation in the development of capabilities and their link to strategy. We consider that the conceptual path connecting CU and ambidexterity has a strong research potential akin to previous empirical studies analyzing ambidexterity in different types of company (Lubatkin, Simsek, Ling, and Veiga, 2006). Moreover, the development of a CU does not occur in a vacuum and is subject to power struggles within the organization and to conflicts between departments which compete for budgets and priorities. Although prior studies have investigated issues of social and intellectual capital (Parshakov and Shakina, 2018), they have overlooked the power issues and conflicts inherent to the development of a CU. Thus, future studies may adopt a practice perspective to provide a sociological understanding of the sources of influence and the resources at stake in the governance (Lissillour and Bonet Fernandez, 2020; Monod, Lissillour, Koester, and Qi, 2022) of a CU. Indeed, the development of a CU may occur at the expenses of other units, which are excluded from strategic affairs such as the development of dynamic capabilities. Finally organizational learning may develop differently across national boundaries (Bogolyubov and Easterby-Smith, 2013; Bogolyubov, Easterby-Smith, and Stead, 2012; Bogolyubov, 2020), consequently further studies may look at cultural difference in the way CU contribute to ambidexterity implementation in different national contexts.

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Appendix

Year	Key event	Developed capacity (code)
2003	The company focuses on internal training, previously primarily focused on external clients	AD04 AD08 AB04 IN013
	The commercial process and portfolio are reorganized. The organizational structure is adjusted	AD01 AD04 AB04 IN02 IN05 IN06
	A planning department was created to plan administrative training and manage three subdivisions	AD01 AD04 AB01 AB02 AB03 AB04 IN01 IN05 IN06 IN08 IN010 IN011 IN012 IN013 IN015 IN016
	A teaching department was created to design and organize technical conferences for clients	AD01 AD04 AB01 AB02 AB03 AB04 IN01 IN03 IN05 IN06 IN08 IN010 IN011 IN012 IN013 IN015 IN016
	A new customer department was created, responsible for administrative affairs and customer relationship management	AD01 AD04 AB01 AB02 AB03 AB04 IN03 IN05 IN06 IN08 IN010 IN011 IN012 IN015 IN016
	A reengineering process is implemented to improve the program quality	AD01 AD04 AD05 AD08 AB02 AB03 AB04 IN01 IN05 IN06 IN08 IN011 IN012 IN014 IN015 IN016
	Different specialists are hired with the necessary skills to design and implement academic and administrative reforms	AD01 AD04 AB02 IN05 IN06 IN08
	An adaptation of the infrastructure is carried out to support the new organization of the university	AD01 AD02 AB04 IN03 IN06
	New resources (teachers, talented customer service staff, and new training programs) are transferred to the internal training division	AD01 AD02 AD04 AB04 IN03 IN06 IN012 IN013
	The top management encouraged the CU to create new training programs targeted to customers in order to improve the marketing of new products	AD02 AD04 AD06 AD08 AB01 AB03 IN06 IN09 IN010 IN011 IN013 IN014 IN016
	The CU incorporates the design and preparation of documentation, which implies the integration of technical and commercial information	AD01 AD02 AD04 AD05 AD07 AD08 AB04 IN01 IN05 IN06 IN09 IN010 IN013 IN014 IN015 IN016
2006	The internationalization of research is promoted through agreements with different partners and the establishment of international headquarters to consolidate a platform for scientific cooperation, and a network to increase sales and deliver services	AD01 AD02 AD03 AD04 AD06 AD07 AD08 AB01 AB02 AB03 AB04 IN01 IN02 IN03 IN04 IN05 IN06 IN08 IN010 IN011 IN012 IN013 IN014 IN015 IN016
	Internally, the company improved the talent of highly qualified staff, facilitating professional exchange with international mobility opportunities	AD01 AD04 AD05 AD07 AB01 AB02 AB03 AB04 IN01 IN03 IN06 IN08 IN010 IN011 IN013 IN015 IN016
	Externally, it extended its training model to clients from different latitudes, achieving an increase in visits and the promotion of new services	AD01 AD02 AD03 AD04 AD08 AB03 AB04 IN01 IN03 IN06 IN08 IN09 IN010 IN011 IN012 IN013 IN014 IN016
	The scope of the service provided was extended which immediately required the training of employees and customers internationally	AD01 AD07 AB01 AB02 AB03 AB04 IN01 IN06 IN08 IN010 IN011 IN013 IN015 IN016
	The creation of new international courses meant the increase of new training capacities, especially linguistic ones, and the incorporation of new teachers	AD01 AD07 AB01 AB02 AB03 AB04 IN01 IN06 IN08 IN010 IN011 IN013 IN015 IN016
	New teaching modalities began to be exploited, taking advantage of the available technologies, the electronic	

(*continued*)

Table A1. Summary of events and capacities developed

Year	Key event	Developed capacity (code)
	learning system was implemented, new multimedia courses and simulation software were created	AD01 AD02 AD03 AD07 AB01 AB02 AB03 AB04 IN01 IN06 IN07 IN08 IN010 IN011 IN013 IN015 IN016
	An English language support department was created	AD01 AD04 AB01 AB02 AB03 AB04 IN01 IN06 IN08 IN010 IN011 IN012 IN013 IN015 IN016
2007	The content of the management training is modified in order to connect the content with the different strategic guidelines of ZTE	AD01 AD02 AD04 AD05 AD07 AD08 AB04 IN01 IN05 IN06 IN09 IN010 IN013 IN014 IN015 IN016
	The participation of top leaders in the design and delivery of content ensured organizational acceptance and that the philosophy and strategic implementation plan were correctly transmitted to the managers	AD01 AD02 AD04 AD05 AD06 AD07 AD08 AB01 AB02 AB03 AB04 IN01 IN03 IN05 IN06 IN09 IN010 IN011 IN013 IN014 IN015 IN016
2008	12 training centers were established in the country where the business volume is important or strategically important	AD01 AD02 AD03 AD04 AD06 AB01 AB02 AB03 AB04 IN01 IN02 IN03 IN05 IN06 IN08 IN011 IN013 IN014 IN015 IN016
	The creation of the project management office (PMO)	AD01 AD02 AD04 AD07 AD08 AB01 AB02 AB03 AB04 IN01 IN02 IN03 IN05 IN06 IN08 IN09 IN010 IN011 IN012 IN015 IN016
2009	Creation of training seminars to promote the quality policy and corporate culture. Its main purpose was to achieve an agreement or consensus concerning the corporate culture	AD01 AD02 AD04 AD05 AD06 AD07 AD08 AB01 AB02 AB03 AB04 IN01 IN03 IN05 IN06 IN09 IN010 IN011 IN013 IN014 IN015 IN016
2010	An alliance and integration of CSL Hong Kong with ZTE is being developed. It created a strong business unit that transferred additional resources to the CU, thus more power and additional capabilities, notably in the area of wireless networks	AD01 AD02 AD03 AD04 AD06 AD07 AD08 AB01 AB02 AB03 AB04 IN01 IN02 IN03 IN04 IN05 IN06 IN07 IN08 IN010 IN011 IN014 IN015 IN016
	Leadership seminars were created that became permanent training called the Leadership Development Program	AD01 AD02 AD04 AD05 AD06 AD07 AD08 AB01 AB02 AB03 AB04 IN01 IN03 IN05 IN06 IN09 IN010 IN011 IN013 IN014 IN015 IN016
	The Leadership Development Program became a program of strategic planning, hosting meetings, agreements and dialectical discussion that led to strategic measures that were later implemented successfully across the enterprise	AD01 AD02 AD04 AD05 AD06 AD07 AD08 AB01 AB02 AB03 AB04 IN01 IN03 IN05 IN06 IN09 IN010 IN011 IN013 IN014 IN015 IN016

Table A1.

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